

A Primer on Social Trading Networks – Institutional Aspects and Empirical Evidence

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Abstract:

Social trading networks provide access to an innovative type of delegated portfolio management. This paper provides a rationale for how these platforms are organized and gives some very first empirical insights to make social trading more tangible for both academics and practitioners alike. First, we discuss the basic mechanics and institutional aspects in light of the arising agency relationships between signal providers (users sharing their investment ideas, i.e. portfolio managers) and signal followers (users subscribing to the signals of other users, i.e. investors). We argue that as an intermediary, the platforms may generally reduce information asymmetries between both groups. However, due to the de facto non-existent entry barriers, it is reasonable to expect a high number of uninformed charlatans among signal providers. Second, using a unique dataset comprising transactions from the four major platforms, we analyze the returns generated during the year 2012. We find that signal providers typically engage in active trading rather than buy-and-hold strategies, which ultimately results in non-normal return distributions, implying that mean-variance analysis is an insufficient framework for performance analysis and portfolio selection in the context of social trading. More precisely, we argue that signal providers typically follow directional approaches and thus may exhibit substantial market risk at any point in time. Hence, far more than the documented 25 percent of the signal providers in our sample can be assumed to bear systematic risk.

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1 Introduction

The financial industry is experiencing an ongoing transformation. Besides the challenges induced by regulatory changes, traditional financial institutions face an increasing competition by innovative startups (*fin techs*) that allow its customers to access conventional financial services by bypassing financial intermediaries in the classical sense. One of these latest innovations is the emergence of online social networks that allow making investment decisions based upon information gathered in online communities, henceforth also referred to as “social trading platforms”.

The recent financial crisis was followed by a general loss of trust in financial institutions and led to a growing willingness among individuals to improve their financial literacy and communicate about financial issues with their peers. While initially mainstream social networks like Facebook and Twitter were increasingly used to share investment ideas, entrepreneurs meanwhile recognized the commercial potential of a direct interface to the user's brokerage accounts. As of 2008, startups – with eToro, ZuluTrade, ayondo and Currensee being among the first – developed specialized social networks that allowed users for an immediate link to their brokerage accounts, with the core of their business model being the generation of order flow for selected partner brokers.

The main feature of these networks is the so-called “copy trading” mechanism that allows for an automated execution of user-generated investment ideas. Basically, there are two user groups: signal providers and signal followers. Signal providers share investment ideas within the community, which they either execute in a virtual portfolio or their real-money brokerage account. Other users can then become a signal follower by subscribing to one or more signal providers. Subsequently, all signals published by the subscribed users are automatically executed into the follower's portfolio, which can also be either a virtual or brokerage account, proportionally and in real-time.¹ Signal providers receive a platform-specific fee for publishing trading signals.

Though followers do not transfer capital into the signal providers' accounts, the latter de facto act as portfolio managers, as they have indirect control over a stake of the signal followers' capital. Hence, social trading networks provide an innovative framework for delegated portfolio management. Indeed, the Financial Conduct Authority (FCA), one of the major regulatory authorities in the UK, recently stated that it considers the automated execution of trade signals within social networks as portfolio management and therefore requires the platform operators for a corresponding authorization.²

¹ Note that some platforms also offer a manual replication, i.e. allow signal followers to manually approve the execution of incoming trading signals. Manual replication can be considered as investment advisory and was the primary service initially offered by the pioneer platforms. However, the fully-automated copy trading' is the predominant form of signal execution now.

² See ESMA Reference 2012/382 and FCA's One Minute Guide „Copy Trading in the Contract for Difference (CFD) Retail Market“, 12/01/2015.

While social trading platforms initially targeted retail investors, they are increasingly getting institutional investors on board as both investors and signal providers. On the one hand, social trading networks may serve as an appropriate sales channel for institutional asset managers to reach out to digital natives, which are currently reaching adulthood and are thus an increasingly relevant client group. For example, as of May 2015, “wikifolio”, one of the more recently founded platforms, already counts roughly 60 institutional asset managers sharing their investment strategies over its website. On the other hand, institutional investors may use these platforms as a fertile ground to screen for new investment ideas or hire portfolio managers.

The aim of this paper is to provide an introduction to social trading networks in order to make them more tangible for both practitioners and academics. Besides discussing the institutional aspects in light of the theoretical and empirical literature on delegated portfolio management (e.g., Stracca, 2006), we draw some very first quantitative insights on the strategies and returns offered by these platforms. Throughout the paper, we compare social trading platforms to mutual and hedge funds as traditional forms of delegated portfolio management.

Specifically, the contribution of this paper is threefold. First, we theoretically outline the role of the platform operator as an intermediary that aims to reduce agency problems and highlight the regulatory guidelines imposed. We argue that while the platforms aim to reduce both the ex ante and ex post resulting information asymmetry, they are likely to attract a high proportion of charlatans. Second, using a unique dataset covering the signals published on the major platforms during the year 2012, we analyze the return characteristics and investment strategies employed. We find that subscribing to a user within a social trading network typically yields strongly non-normal returns and exposes the investor to considerable tail risk. In our sample, only the top 50 percent of signal providers with the highest return are able to keep up with mutual funds net of fees. To classify the investment strategies prevailing in social trading networks, we then perform a return-based style analysis following Sharpe (1992) and Fung and Hsieh (1997). We find that signal providers rather deploy dynamic trading than buy-and-hold strategies. More precisely, we argue that they typically follow directional approaches and thus may exhibit substantial market risk at any point in time. Hence, far more than the documented 25 percent of the signal providers in our sample can be assumed to bear systematic risk.

The remainder of this paper is organized as follows. In the following section, we describe the institutional framework of social trading networks. We outline how the platform operator acts as an intermediary and explain how they are regulated. In section 3, we first examine the return characteristics and trading strategies accessible by subscribing to a signal provider. We then analyze the systematic risk and directionality involved with these strategies. Section 4 summarizes our results and provides some concluding remarks.

2 Institutional Characteristics of Social Trading

2.1 Agency Problems and Incentive Structures

While signal providers de facto manage the followers' brokerage accounts, an immediate legal relationship solely exists between the two user groups and the social trading network as an intermediary. Besides routing trading signals, the platforms aim to mitigate the potential agency problems inherent to any delegated portfolio management setting (see the seminal paper by Bhattacharya and Pfleiderer, 1985).

First, they attempt to reduce the ex-ante (i.e. the before subscribing to a user existing) information asymmetry by publishing a full single transaction history and standardized real-time track records for each of the signal providers signed up to the platform. These are further extended by several screening instruments such as (sometimes pre-configured) rankings and search functions. Compared to mutual and hedge funds as the most popular traditional forms of portfolio management, social trading platforms thus provide a greater depth of information on a manager's historical performance. Hedge funds usually do not disclose portfolio holdings at all (e.g., Fung and Hsieh, 1999), while most mutual funds report holding data only on a quarterly or semi-annual frequency (Elton et al., 2010). However, as discussed by Huddart (1999), detailed information on historical trades alone does not guarantee that investors succeed in assessing whether a good past performance is due to skill or just the result of a lucky coincidence, as they typically do not know the process behind what indicators triggered these transactions (unless the signal provider voluntarily provides any additional and likewise reliable comments on his or her decision). Thus, the high level of transparency does not per se reduce the ex-ante information asymmetry.

We assume that despite the depth of the track records provided, isolating skilled portfolio managers is a task that is particularly difficult on social trading platforms. The reason is that basically any internet user is eligible to join such a network and begin publishing trading signals, i.e. there are no considerable admission requirements. As Huddart (1999) points out, portfolio manager markets with low entry barriers are prone to attracting charlatans, that aim to build a solid track record by solely betting on lucky chances. Thus, while the high transparency provided by detailed track records might itself be somewhat helpful to investors, low entry barriers make it reasonable to assume a particularly high number of charlatans on social trading platforms. Thus, when attempting to filter out skilled signal providers, investors have to deal with a lot of "noise". It is hence unclear whether users on social trading networks can theoretically make better manager selections than among more traditional forms of delegated portfolio management.

In every agency-relation, the principal bears some risk that ex post, i.e. after selecting an agent, the latter may be engaging in actions that are not in the principal's best interest (moral hazard). A special feature of social trading platforms here is that after subscribing to a signal provider, the subsequently shared

investment signals are automatically mirrored into the investor's brokerage account. Compared to mutual and hedge funds, where investors can typically only observe daily and monthly changes in net asset values, respectively, they can track the trading activities on social trading platforms in real-time. Hence, social trading platforms facilitate the monitoring of managerial actions, thus reducing the ex-post information asymmetry compared to traditional forms of portfolio management. Knowing that investors can ex post monitor their trading activities in real time, signal providers might be less tempted than under a traditional delegated portfolio management setting to engage in moral hazard.

Theoretically, platforms could reduce moral hazard by the choice of compensation structure. In general, the compensation contract is considered to play a pivotal role in aligning both parties' interests in any principal-agent relationship. It should be chosen in a way that ensures the agent undertakes his best efforts to serve the principal's interests. However, in this context, delegated portfolio management settings differ in one important aspect: the agent does not only decide about how much effort to take to generate informed investment ideas, but compared to more common agency-relations also has a more direct control over the scale of his decisions to a gathered signal, i.e. over how much risk he takes (Stracca, 2006). There is still an ongoing debate in the literature on how to optimally compensate portfolio managers, as compensation contracts should not only motivate managers to undertake sufficient effort, but also avoid providing her an incentive to take excessive risk (see Stracca, 2006 for a good survey on this discussion). Thus, it is important to note here that there is no consensus on a single optimal compensation model for portfolio managers yet.

Among more traditional forms of delegated portfolio management, two different compensation schemes are predominant. First, mutual funds are subjected to the *fulcrum rule*, implying that gains and losses must have a symmetric effect on their fees. In other words, they are not allowed to impose non-linear fees. Thus, mutual funds typically only charge management fees, i.e. a fixed percentage share of their assets under management. Hedge funds, on the other hand, are exempted from this rule, enabling them to impose convex fees, i.e. fees that reward a positive performance, but do not require them to rebate fees to investors in case of losses (Fung and Hsieh, 1999). Hence, the hedge fund industry standard is to charge both a management and performance fee, with the latter typically being tied to a "high-water mark" (henceforth: HWM) provision. HWM provisions condition the payment of the performance fee on net profits exceeding a prior maximum net asset value, i.e. require the fund to make up previous losses before being allowed to charge performance fees (e.g., Aragon and Nanda, 2012). For social trading platforms, a customary compensation model has not evolved yet. Yet, a rough distinction can be drawn between three groups: follower-based models, profit-based models containing a HWM, and volume-based models.

A *follower-based* model pays signal providers a fixed amount per subscription. For example, eToro, the largest provider by users as of July 2015, pays a lump sum depending on the number of relevant followers, limited to a monthly payment of \$20,000. Followers are defined as being “relevant” when they assigned an amount of at least \$100 to that signal provider. As signal providers are incentivized to maximize the number of investors, a follower-based fee is most similar to the asset-based fee regularly charged by mutual funds. Since asset-based fees are proportional to the assets under management, they should theoretically reward gains and penalize losses to similar extents. However, Sirri and Tufano (1998) and Chevallier and Ellison (1997) document that among mutual funds, investors react disproportionately to a good and bad prior performance. In other words, mutual fund flows react considerably stronger to gains than to similar losses, effectively limiting portfolio managers’ liability for losses and exposing them to an implicitly convex fee. Thus, a large number of papers raised concerns whether asset-based fees still motivate portfolio managers to take inappropriate risk in some circumstances (e.g., Brown, Harlow, and Starks, 1996).

In a standard agency setting, a *profit-based* compensation model (i.e. performance fees) that provides explicitly convex incentives has been found to be the optimal approach when information is asymmetrically distributed before and after contracting (see e.g. Holmstrom and Milgrom, 1987; Sappington, 1991). However, it is not considered to be optimal in a delegated portfolio management setting (e.g., Stracca, 2006). In general, when subjected to a performance fee, a portfolio manager is effectively holding a call-option on investor wealth. As analytically shown by Carpenter (2000), this can induce fund managers to take excessive risk when they are far out of the money. Therefore, it became a popular practice among hedge funds, who typically derive a substantial part of their income from performance fees, to tie their payment to a HWM provision (Aragon and Nanda, 2012). Among social trading networks, especially more recently founded platforms such as “wikifolio” and “United Signals” follow this approach.

Under a HWM provision, taking excessive risk on the one hand increases a signal provider’s chance of crossing the HWM and realize a performance fee. On the other hand, it also increases her risk of starting off future periods with an even lower moneyness. Thus, as shown analytically by Panageas and Westerfield (2009) and Hodder and Jackwerth (2007), if the time horizon is long enough, the imposition of a HWM can indeed somewhat reduce the incentive to take excessive risk compared to a plain performance fee without HWM provision. This is also supported empirically by the findings of Aragon and Nanda (2012). However, despite being *lower* compared to performance fees without HWM, they find that hedge fund managers are still prone to game the convexity of their incentives by taking excessive risk.

With the development of social trading networks, a compensation model that is so far unusual in the context of delegated portfolio management emerged: *volume-based* fees. Here, signal providers receive a rebate on the overall trading volume generated by their published trading signals, i.e. a share of the

respective bid-ask spread. For example, “ZuluTrade” rebates their signal providers a share of approximately 0.005 percent of the generated volume. Thus, it does not distinguish between profitable and non-profitable signals. While this may prevent signal providers from holding on to losing positions for too long in order to maintain their chance for receiving a fee, such a compensation scheme may intuitively increase the risk of churning by randomly opening and closing out positions. Therefore, some networks (e.g., “ayondo”) scale the magnitude of the percentage share rebated to a signal provider by her historical performance.

Finally, two things should be noted here. First, the large variety of different compensation models among social trading platforms is unsurprising, given that the literature on delegated portfolio management failed to derive a single best contract yet (e.g., Stracca, 2006). This makes it impossible to ultimately argue in favor or against one of the models discussed above, but also casts doubt on the ability of the platforms to mitigate the discussed agency problems solely by their choice of compensation. Second, since social trading platforms are two-sided markets, the choice of compensation scheme is also of particular importance for the commercial success of the platform operators themselves. As talented signal providers attract additional investors and vice versa, both user groups induce positive external effects (see Rochet and Tirole, 2003).

2.2 Tradable Instruments and Regulatory View

Typically, social trading platforms solely offer over the counter (OTC) trading with so-called *contracts for difference* (CFD). A CFD is a short term total return swap on the returns of an underlying asset versus an interest rate, i.e. the buyer of such a swap receives the difference between the current value of an underlying asset and its value at contract time and pays a (fixed) interest to the seller (e.g., Alexander, 2008). The seller, on the other hand, makes a profit when the underlying value depreciates. Thus, with CFDs, investors can bet on both rising and falling prices likewise. Since CFDs are traded on margin, both parties solely deposit a cash collateral and therefore participate disproportionately in price movements of the underlying asset. Unlike futures, CFDs do not have a fixed expiry date or contract size. The high flexibility in terms of contract sizes allows for a fractional mapping and hence to ensure an exact proportionality between the signal provider's and followers' accounts (Doering, Heyden, Surminski, 2013). Since the counterparty (i.e. the broker) usually closes out the position in case of a margin call, CFDs are solely offered on underlyings with an appropriate market depth and liquidity. Thus, the investment universe on social trading platforms is typically constrained to foreign exchange (forex) pairs, equity indices and major single stocks, commodities and bond indices. The set of assets tradable by signal providers is therefore solely a subset of the total investment universe accessible by more established forms of delegated portfolio management (such as mutual or hedge funds).

For member states of the European Union (EU), the European Securities and Markets Authority (ESMA) announced in 2012 that it considers social trading platforms as exercising “investment discretion by

automatically executing the trade signals of third parties”, therefore requiring them to have an authorization corresponding with Markets in Financial Instruments Directive (MiFID). Hence, within the EU, the platform operators themselves instead of the signal providers are assumed to be portfolio managers from a regulatory point of view. The FCA recently announced that it supports this view.³

3 Empirical Analysis of Signal Provider Returns

3.1 Data

Our sample comprises the 2012 track records of signal providers with at least 50 followers as of December 31, 2011, registered on one of the four major platforms, namely eToro, ZuluTrade, ayondo and Currensee. In total, this amounts to a sample of 150 signal providers, with 52 for eToro, 76 for ZuluTrade, 14 for ayondo and 8 for Currensee. The track records provide a full history of published trade signals, including the traded underlying, trade direction (long/short), leverage, buy and sell prices, as well as the date and time for each signal.

We converted these transaction histories to weekly portfolio returns, resulting in 52 returns for each signal provider and to 7,800 observations in total. The initially computed portfolio returns considered profits and losses of a position only after it was closed out. In contrast, the returns of an investment fund typically include unrealized gains and losses, as the net asset values account for price movements in open positions. In order to ensure comparability to traditional asset managers, we thus artificially closed all open positions at the end of a trading week and reopened them at the beginning of the subsequent trading week. To compute returns of pending trades, we obtained the opening and closing prices of the underlying securities from Thomson Reuters Datastream. Since the bid-ask spread charged by social trading platforms does not only account for trading fees, but also includes a share for signal provider compensation and the platform operators’ profit margin, the calculated returns are net of all fees.

For our later return-based style analysis, we follow the asset factor selection of Fung and Hsieh (1997) and employ three equity classes, proxied by the following indices: MSCI US equities, MSCI non-US equities, and MSCI emerging market equities. We use three bond indices: iBoxx eurozone sovereign bonds, iBoxx eurozone corporate bonds, and Datastream US government bonds. For cash we use the three-month Euro Interbank Offered Rate (Euribor) and for commodities the price of gold (LBMA fixing). Finally, currencies are represented by the Euro Currency Index. For all factors considered, we sourced total returns from Thomson Reuters Datastream. In addition, for the purpose of comparison with traditional forms of portfolio management, we also apply the factors above to mutual funds, which we proxy by using the total returns of the 35 Lipper Average indices that represent the major mutual fund

³ See ESMA Reference 2012/382 and FCA's One Minute Guide “Copy trading in the contract for difference (CFD) retail market”, 12/01/2015.

investment styles. These indices are calculated in a way that they are net of management and performance, but gross of trading fees. Weekly total returns for the Lipper Average were also retrieved from Thomson Reuters Datastream.

3.2 Summary Statistics and Downside Risk

Table 1 reports summary statistics for signal provider, mutual fund and asset factor returns. For signal providers and mutual funds, we first compute each measure separately for the return time-series and then report the cross-sectional median. Obviously, with a weekly mean return of a mere 0.07 percent for the full sample and a standard deviation of 1.79 percent, the median signal provider did not outperform the median mutual fund or any of the market indices considered during 2012. However, when we divide the overall sample into those with an above- and below-median return, we at least observe the top signal providers achieving a performance that is similar to mutual funds. Both average returns and return standard deviations are twice as large as for the typical mutual fund. On the other hand, signal providers with a below-median profitability exhibit a zero-average return.

[Insert Table 1 here]

The results of the Jarque-Bera test (Bera and Jarque, 1980) show that in contrast to mutual funds and the reported market indices, the majority of signal providers generates returns that are significantly non-normally distributed. This holds true for both the full and sub samples. Their return distributions are characterized by a negative skewness and a, compared to mutual funds and other asset classes, substantial excess kurtosis of roughly two. Thus, care should be taken when applying mean-variance analysis to evaluate signal providers' performance: the commonly underlying assumption of normally distributed returns may lead to an overestimation of true performance (e.g., Lamm, 2003).

It is widely accepted that a combination of negative skewness and large kurtosis is opposite from what investors favor. For example, Scott and Horvath (1980) show that investors have a preference for high (i.e., positive) skewness and low kurtosis. Compared to a normal distribution, negative skewness in combination with positive excess kurtosis implies a substantially higher risk of relatively large losses. Table 2 reports tail risk measures for signal provider and mutual fund returns.

[Insert Table 2 here]

It can be seen that signal providers with a below-median return (that achieved a zero-return anyway, see Table 1) expose investors to a consistently lower tail risk than mutual funds. This does however not hold when measuring downside risk across the entire cross-section of signal providers or focusing only on those with an above-median return. Then, we observe signal providers exhibiting both a higher plain and conditional value-at-risk (VaR). For example, with a probability of 95 percent, the typical signal provider does not suffer a weekly loss exceeding 2.78 percent. If she does, however, the losses average to 3.85 percent. In comparison, the median mutual fund exhibits a VaR (conditional VaR) of only 2.46

percent (3.59 percent) over the same period. However, when measuring tail risk using maximum and average drawdowns, only signal providers with an above-median return exceed the figures realized by mutual funds and thus exhibit a higher downside risk. As a whole, signal providers do not expose investors to a higher maximum or average drawdown.

Note that signal providers in both the full and the subsamples offer relatively favorable drawdown and recovery durations. For the typical signal provider, drawdown periods last for an average of 7 weeks, and it afterwards takes them on average 3 weeks to recover from the previous losses, i.e. reach a new high-water mark again. In contrast, the average drawdown duration of a typical mutual fund amounts to roughly 9 weeks and the time until recovery to 4.5 weeks. Still, the overall picture emerges that signal providers, especially those that are profitable, expose investors to significantly non-normal returns and a higher tail risk than mutual funds.

Within the framework of delegated portfolio management, non-normal return distributions are frequently associated to dynamic trading strategies, i.e. strategies that in contrast to buy-and-hold approaches make use of short sales and regularly lever their bets by margining or employing derivatives (e.g. Agarwal and Naik, 2004; Gregoriou and Gueyie, 2003). In fact, portfolios implementing these kinds of strategies often exhibit option like return patterns (see Glosten and Jagannathan, 1994; Mitchell and Pulvino, 2001; Fung and Hsieh, 2001).

3.3 Return Determinants and Trading Strategies

A statistical technique frequently used to classify the investment strategy of a portfolio manager is the return-based style analysis introduced by Sharpe (1992), who proposed a multi-asset factor model to estimate the portfolios' exposures to different asset markets. In order to examine return determinants and to answer the question whether signal providers rather deploy dynamic trading strategies or buy-and-hold strategies, we follow Sharpe (1992) and estimate the model below separately for all signal providers and mutual funds in our sample:

$$r_{t,m} = \alpha_m + \sum_{k=1}^K \beta_{k,m} r_{t,k} + \epsilon_{t,m}, \quad t = 1, \dots, T, \quad (1)$$

where $r_{t,m}$ is the return of signal provider or mutual fund index m over week t , and $r_{t,k}$ the return of factor k over week t . As factors, we use the eight indices outlined in section 3.1, a selection which follows the one used by Fung and Hsieh (1997) to classify hedge fund strategies. From the estimates of the slope coefficients β_k we can infer the average exposures of the signal providers' and mutual funds' strategies to certain security markets.

In our baseline regression, we also impose the following constraints following Sharpe (1992):

$$\sum_{k=1}^K \beta_{k,m} = 1 \quad (2)$$

$$\beta_{k,m} > 0, \quad \forall k. \quad (3)$$

Equation 2 prohibits the use of leverage, whereas equation 3 forbids short sales. In order to analyze whether short sales and leverage are indeed important factors contributing to signal providers' returns, we then remove the above constraints step-by-step. Thus, we first perform a regression subjected to both of the above two conditions. We estimate this constrained regression using quadratic programming. Then, we carry out a regression that allows for using leverage, but still prohibits short sales. Finally, we run a completely unconstrained regression, i.e. estimate the model in equation 1 without imposing any further conditions on the slope coefficients.

[insert Table 3 here]

Table 3 reports the distribution of R^2 s for the three regressions separately for signal providers and mutual funds. The results show that for mutual funds, the distribution remains almost unchanged when removing leverage and short-selling conditions – the constrained regression fits the data nearly as good as a completely unconstrained model. These results are qualitatively and quantitatively very similar to those reported in Sharpe (1992).

However, these results substantially differ for signal providers. First, the overall fit is considerably smaller. While for mutual funds, the fully-constrained model in no case explains less than 55 percent of the time-series return variation, we observe negative R^2 s for at least 25 percent of the signal providers in our sample when applying the same model.⁴ Allowing for short sales slightly increases the fit, but R^2 s are still below 0.20 for at least 75 percent of all signal providers in our sample. However, if we additionally remove the leverage constraint, values significantly increase. For example, the median R^2 s increase by a factor of 13 from 0.02 to 0.26.

Since the stepwise removal of trading limitations does virtually not increase the fit for mutual funds, their returns are evidently largely driven by the asset allocation choice. Leverage and short sales are, if any, minor return drivers. Hence, in the words of Fung and Hsieh (1997), the typical mutual fund strategy is mostly about *where* and much less *how* they invest – given the choice of markets, mutual funds' strategies are quite similar to buy-and-hold. In contrast, as the fit to signal provider returns considerably increases, leverage and short sales seem to be integral parts of the strategies prevailing on social trading platforms.

Unlike for mutual or hedge funds, the high transparency of these platforms reflected in our comprehensive data set allows us to observe the direction and leverage of each single trade signal published by a user. This enables us to analyze whether the majority of signal providers really makes frequent use of

⁴ Note that, while a negative *adjusted* R^2 is not rare in regressions, the plain R^2 can solely be negative for constrained regressions. This is the case if constraints enforce an estimation that fits the data worse than a horizontal line, i.e. the mean of the dependent variable provides a better explanation for the outcomes than the estimated model (see e.g. Cameron and Windmeijer, 1997).

margin and short sales. Table 4 shows descriptive statistics on the trading behavior of signal providers, reported separately for each network covered by our sample.

[Insert Table 4 here]

Obviously, as the mean share of long trades ranges between about 50 and 60 percent across all four platforms (Panels A to D), the majority of signal providers indeed bets on both rising and falling markets dynamically. However, note that as the maximum long share for eToro and ayondo is close to 100 percent, there are also a few individuals that can be considered as long-only traders.

Though the average leverage is above one for all four platforms, indicating that signal providers indeed typically lever their signals by margin trading, there is also considerable variation across the platforms. We observe relatively large leverages of roughly three and nine for ZuluTrade and ayondo, respectively, and average leverages below two for eToro and Currensee. A potential explanation for this pattern is the volume-based fee paid to signal providers on ZuluTrade and ayondo. This may motivate them to suggest trading signals with a higher leverage, as greater leverage implies more trading volume and thus a greater compensation.⁵ In line with this rationale, we also observe a higher trading frequency among these two platforms. However, even for Currensee and eToro we see an average of approximately 10 trades per week for the typical signal provider. Altogether, these observations strongly suggest that the majority of signal providers indeed follows active, dynamic trading rather than buy-and-hold strategies.

3.4 Market Neutrality of Signal Provider Returns

It should be noted that even after removing leverage and short sale constraints, the overall explanatory power of the style regressions reported in Table 3 is still quite low for signal providers, compared to mutual funds. The fit of the unconstrained regressions for both signal providers and mutual funds are illustrated in more detail Figure 1. It can be seen that the R^2 s for the majority of signal providers are below 0.4, while mutual funds accumulate above 0.5. The observation that common security markets appear to have low power to explain signal provider returns gives some initial evidence that the returns offered by social trading platforms are largely market neutral.

[insert Figure 1 here]

Market neutrality is one of the most frequently mentioned features of dynamic trading strategies (e.g., Agarwal and Naik, 2004; Goetzmann et al., 1999). In fact, the diversification benefits of hedge funds, which are the most popular market participants engaging in dynamic trading strategies, are one of their most-cited selling points (e.g., Patton, 2009). In order to examine whether social trading networks indeed

⁵ On eToro, signal providers are compensated with a fixed remuneration, determined by the number of followers and limited to 10,000 US dollar per month. We assume the compensation cap to provide an incentive for “locking in” a given income level and hence reduce excessive risk taking. On the other hand, signal providers operating on Currensee receive a performance-based fee including a high-water mark and therefore solely participate in net profits. Panageas and Westerfield (2009) outline that such a contract can constrain excessive risk-taking.

offer market neutral returns, we perform overall- F -tests based on the the unconstrained regressions discussed above, which test on the joint irrelevance of all eight asset markets to explain a signal provider's or mutual fund's returns.

[Insert Table 5 here]

Table 5 reports the distribution of p -values for these tests, separately for both mutual funds and signal providers. The results show that the hypothesis of market neutrality must be rejected for every single mutual fund in our sample. On the other hand, it cannot be rejected for at least 50 percent of the signal providers. Thus, at least half of the signal providers in our sample generate returns that are not significantly related to any common security market. Figure 2 presents the percentage shares of signal providers and mutual funds that are significantly exposed to the corresponding factors, based on t -tests and a one-percent level of significance. It can be seen that the vast majority of mutual funds is related to US and non-US equities (about 70 and 80 percent, respectively). A considerable share of about 40 percent also exhibits a significant exposure to European government bonds. In contrast, for any of the factors considered, the percentage share of signal providers that have a significant exposure is always below 20 percent. The above findings generally support the view that trading strategies offered on social trading platforms are mostly market neutral, suggesting that they may be a good diversifier in a balanced portfolio of more common assets.

[Insert Figure 2 here]

However, in analyzing their systematic risk exposures, we have so far treated signal providers as a whole. Yet, considering that they do not face considerable investment guidelines, it is reasonable to assume that they engage in a variety of different investment styles, which ultimately come along with different amounts of market risk. In the context of hedge funds, Fung and Hsieh (1999) argue that investment style should be thought of in two dimensions: (i) the choice of asset classes to invest in, and (ii) the trading strategy, i.e. the decision on how to transact in these asset classes. Specifically, “trading strategy” on the one hand refers to the direction (long or short) – i.e. the sign of the slope coefficients $\beta_{k,m}$ in style regressions – and, on the other hand, to the leverage employed, which affects the magnitude of slope coefficients. Hence, the actual returns are the products of allocation choice and trading strategy (see Fung and Hsieh, 1997).

Regarding the location choice, signal providers are somewhat more limited than mutual funds, since the set of tradable assets on social trading platforms is only a subset of the total investment universe accessible by mutual funds. On the other hand, they are more flexible with respect to the trading strategy, since they can vary both the direction and leverage of their positions. As discussed in Fung and Hsieh (1999), mutual funds typically must comply with strict investment guidelines as defined in their prospectuses, which forbid making extensive use of margin trading, derivatives or short sales. Hence, the

observed differences in market neutrality between mutual funds and signal providers most likely stem from the way in which the latter transact in securities, i.e. their trading strategies.

Thus, in what follows, we analyze the extent to which the documented market neutrality of signal providers' returns varies with their trading strategies. Following the above logic of Fung and Hsieh (1999), we examine whether there are any differences in market neutrality for signal providers that (i) frequently change the direction (long/short) of their trades, and (ii) that often vary their leverage employed.

Specifically, we first regress the R^2 s of the unconstrained style regressions from section 3.3 on the corresponding percentage shares of long trades of the signal providers. To allow for a non-linear relation, we do this by performing a second-degree polynomial regression, i.e. we model R^2 s as a 2nd degree polynomial of the share of buy trades. To assess whether a higher variation in the leverage used contributes to a lower fit, we additionally divide the data into two subsamples, based on whether the signal provider has an above- or below-median leverage variation. Since the simple standard deviation is likely to be higher for portfolios that generally maintain a higher leverage level, we use the median of the variation coefficient, i.e. the ratio between leverage standard deviation and mean leverage, as a cut-off value. We then also run the above polynomial regression separately for each subsample.

Figure 3 plots the results for this regression. The solid black line shows the estimated curve for the full sample, while the dashed (dotted) line plots the curve for signal providers with a low (high) variation in leverage. First, it can be seen that the explanatory power of security market returns to explain signal providers' returns varies strongly with the ratio of long and short trades. R^2 s are relatively high for signal providers that primarily trade in a single direction, and are lowest for those with a balanced ratio of long and short positions. Besides, we observe that at least for signal providers that mostly trade either long or short, explanatory power also decreases for those users that frequently vary the leverage of their positions: the higher the variation in the magnitude of leverages, the lower the R^2 s. However, the impact of leverage variation is negligible for signal providers that frequently vary their trade direction anyway.

[Insert Figure 3 here]

In Figure 4, we repeat the above analysis, but this time using the p -value of the overall F -tests instead of R^2 s as dependent variable. Corresponding to the results in Figure 3, we now observe an inverse U-shaped relationship between p -values and the percentage share of long trades. Thus, the documented market neutrality strongly depends on the ratio of trade directions. We observe market neutrality especially among those signal providers who rather engage in long-short strategies. On the other hand, signal providers that mostly trade either long or short do not tend to generate market neutral returns. Besides, at least for signal providers with a share of long trades below 60 percent, i.e. those that rather follow long-short or pure short strategies, p -values also appear to be related to leverage variation. Among these

groups, signal providers that use a constant leverage carry a somewhat larger systematic risk than those who frequently vary their leverage from trade to trade.

Altogether, these results suggest that market neutrality strongly depends on the trading strategy employed. Though we found evidence that most signal providers carry no significant systematic risk (see Table 5), when considering to invest via social trading platforms, investors are well-advised to evaluate the correlation of signal providers' returns with market returns on a case-by-case basis.

Consider another cautionary note here. Dynamic trading strategies can either follow directional or non-directional approaches (Fund and Hsieh, 1999). *Non-directional* approaches aim to exploit market anomalies by being long and short in comparable securities at the same time. Consequently, non-directional strategies carry relatively little systematic risk. *Directional* approaches, on the other hand, bet on the direction of markets dynamically by being either long or short, and thus carry considerable systematic risk at any time. As Fung and Hsieh (1999) point out, non-directional strategies often involve holding rather complex securities, such as convertible bonds, and are rather seldom implemented in more common security markets. However, signal providers can CFDs are typically only offered on highly liquid underlyings (see section 2.2). We thus assume that the majority of signal providers rather follows directional approaches. In fact, in unreported results, visual inspection unveiled that even signal providers with a quite balanced ratio between long and short trades tend to follow directional approaches, as they only seldom hold similar amounts long and short at the same time.

In this light, it should be noted that there is controversy in the hedge fund literature on the extent to which dynamic trading strategies are really market neutral (see e.g. Patton, 2009; Ennis and Sebastian, 2003). To illustrate this point, Fung and Hsieh (1997) cite the example of a manager solely trading S&P500 futures contracts. They argue that in a mutual fund (i.e. buy-and-hold) setting, a "fully invested position of being consistently long one futures contract" will result in a style-regression coefficient of +1 on the S&P500 index. Conversely, a hedge fund manager that is fully and consistently invested in a short position will exhibit a coefficient of -1. If, however, the manager alternates between being completely long and short each day, common empirical asset pricing models, such as the linear style analysis applied in this article, will indicate an exposure that is close to zero, even though substantial market risk is taken at any point in time. Consequently, R^2 s for style regression will be low, and F -tests will indicate the manager's returns are unrelated to any of the asset factors considered. This issue stems from the fact that common asset pricing models, such as the CAPM or multi-factor models, assume a linear relation between asset and factor returns, but the returns of dynamic trading strategies are non-linearly related to underlying asset returns.

The bottom line of the above discussion is twofold. First, the share of market neutral signal providers discussed above must be interpreted with caution. Considering that signal providers most likely follow directional strategies, it is reasonable to assume that a large proportion only exhibits seemingly market

neutral returns, because varying exposures over time average out to zero on the observation date. The share of signal providers offering truly market neutral returns can hence be expected to be several magnitudes smaller than the documented 50 percent. Second, care should be taken when applying common asset pricing models to study signal providers' risk exposures. The above results once again suggest that investors are required to apply more sophisticated (i.e. non-linear) models to properly analyze investment strategies offered via social trading platforms.

4 Conclusion

Social trading platforms allow investors to subscribe to one or more signal providers, whose subsequently shared trading signals are then executed within the investor's brokerage account proportionally and in real-time. For sharing their trading ideas, signal providers are paid a fee by the platform provider. Hence, social trading networks provide an innovative form of delegated portfolio management. They experience an increasing popularity among retail and most recently also institutional investors.

Besides providing the technology for trade mirroring, the platforms act as an intermediary between investors and signal providers, aiming to reduce ex-ante and ex-post information asymmetry by providing detailed track records, rankings and search functions, and also allowing investors to track their portfolio managers' trading activities in real-time. On one hand, this allows to perform screening and monitoring activities at a level of transparency that is unique compared to more traditional forms of delegated portfolio management, such as mutual or hedge funds. However, investors should be aware that since there are virtually no barriers to create an account and start sharing trading signals, a high number of charlatans might be tempted to join social trading platforms and try to build up a solid track record solely by making random, high-risk bets. Thus, on the other hand, investing on these platforms may require a particularly extensive screening and monitoring compared to other forms of delegated portfolio management. This holds in particular as signal providers are, unlike mutual and many hedge funds, not subjected to any investment guidelines that limit their exposure to certain asset classes or the size of single positions. While in more general agency settings, both adverse selection and moral hazard can theoretically be mitigated by selecting an appropriate compensation contract, there is still an ongoing debate in the literature on the optimal payment model in delegated portfolio management settings (e.g., Stracca, 2006). In this light, it is not surprising that we observe a large variety of different compensation models among the competing social trading platforms, and it is likely to assume that the compensation model alone will not solve the discussed agency problems.

Using a sample of signals published on major social trading networks during 2012, we show that the returns achievable by subscribing to a signal provider follow a significantly non-normal distribution and exhibit a relatively high tail risk compared to mutual funds. Thus, practitioners considering to invest via social trading platforms are well-advised to employ more sophisticated models for performance analysis

and portfolio selection, as traditional mean-variance analysis fails to properly deal with non-normal return distributions (e.g., Lamm, 2003).

We find that a traditional return-based style analysis according to Sharpe (1992) and with an asset class selection following Fung Hsieh (1997) results in a high fit for mutual funds, even if we constrain for leverage and short sales. In contrast, R^2 s are relatively low for signal provider returns and considerably increase when removing short sale and leverage constraints step-by-step. Thus, short exposures and margin trading appear to be integral parts of the strategies offered on social trading platforms. By analyzing the direction and leverage used for individual signals, we show that signal providers indeed deploy dynamic trading rather than buy-and-hold strategies. On average, between 40 and 50 percent of the signals executed are short trades. Additionally, signal providers typically lever their bets by a factor roughly ranging from 2 to 9.

In line with the lower overall explanatory fit of the style regressions for signal providers, we find that at least half of them exhibit market neutral returns, based on overall F -tests on the joint irrelevance of the asset factors considered. We document market neutrality especially among those signal providers that frequently vary between long and short trades. On the other hand, every single mutual fund in our sample is significantly exposed to at least one asset factor and thus carries systematic risk.

However, the observed market neutrality for a large part of signal providers should be interpreted carefully. It is reasonable to assume that they mostly engage in directional strategies, and thus market neutrality may often only be documented because varying directional long and short exposures simply average out over time, a phenomenon that is also widely discussed in the hedge fund literature (e.g., Ennis and Sebastian, 2003). Thus, measuring systematic risk exposures among signal providers on social trading platforms may require investors to apply more sophisticated concepts than linear style regressions (e.g., Patton, 2009).

Altogether, social trading platforms offer a unique delegated portfolio management framework that combines dynamic trading strategies (that where traditionally associated with hedge funds) with an almost greatest possible transparency. This provides an interesting setting for both investors looking beyond passive strategies as well as future academic research. However, investors should be aware that common analytical tools, such as mean-variance analysis or linear style-regressions, may fail to properly account for the risks inherent to the dynamic strategies offered via these platforms.

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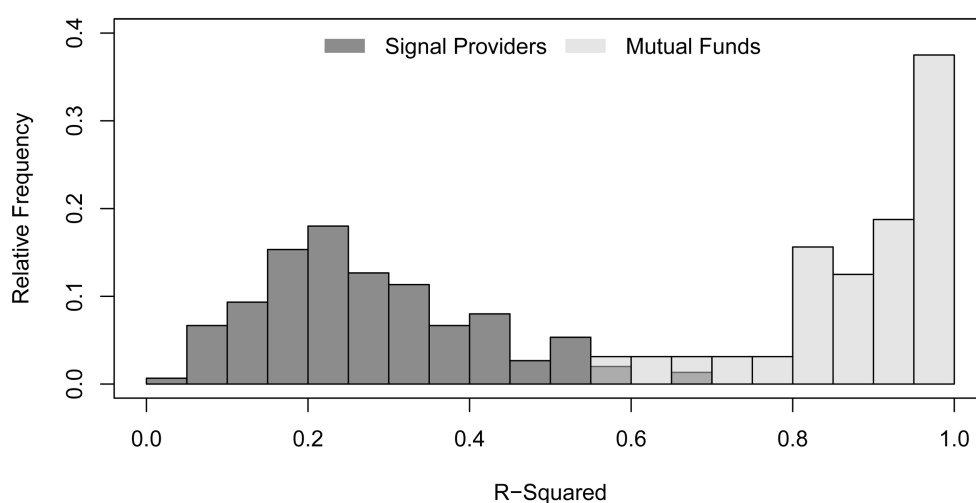
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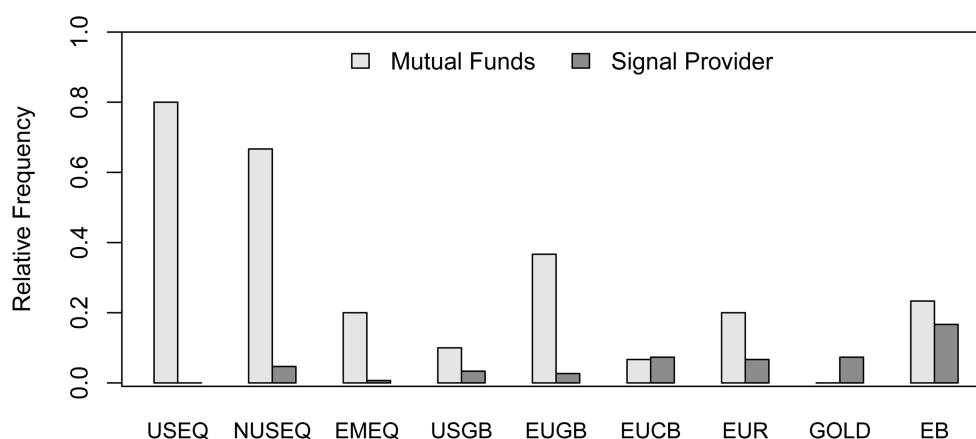
Appendix 1 – Figures

Figure 1 – Cross-Sectional Distribution of R^2 s for Style Regressions



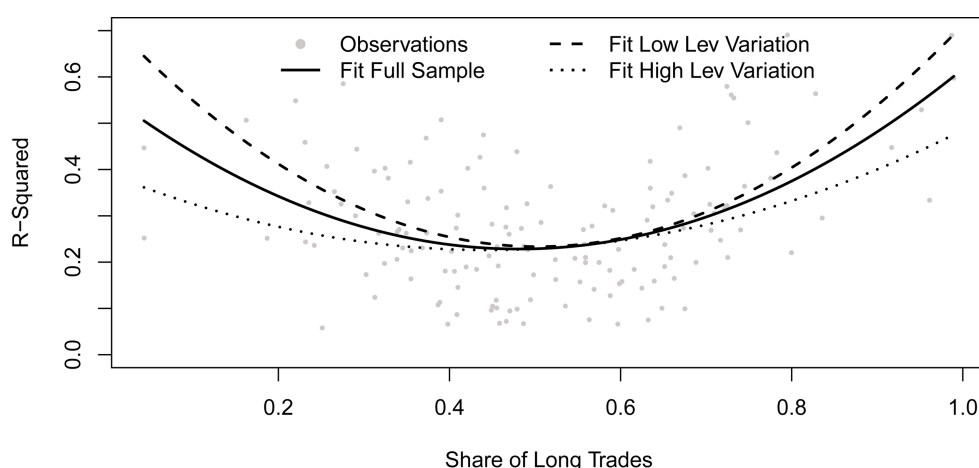
This figure plots the relative frequency of R^2 s for unconstrained style regressions as defined in Equation 1, separately for both signal providers and mutual funds. Regressions are based on weekly returns over the year 2012. The explanatory variables considered are outlined in section 3.1.

Figure 2 – Distribution of Significant Factor Exposures



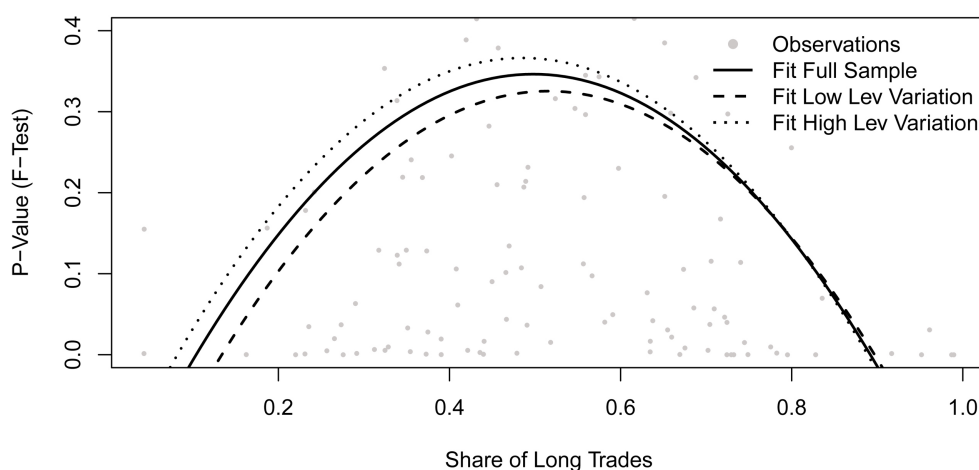
This figure shows the proportion of mutual funds and signal providers with significant exposures to the eight asset factors considered as explanatory variables in our style regressions (see section 3.3). Exposures are considered as being significant when the slope coefficient is different from zero at a one-percent level of significance according to two-sided t -tests. USEQ, NUSEQ and EMEQ refer to US, non-US and emerging markets equities, respectively. USGB, EUGB and EUCB relate to US government, European government and European corporate bonds, respectively. EUR denotes the Euro Currency Index and EB the Euribor (see section 3.1).

Figure 3 – R^2 s for Style Regressions versus Trade Direction and Leverage



This figure shows how R^2 s of the unconstrained style regressions for signal providers vary with their percentage share of signals that imply a long position. We compute the proportion of long signals for a signal provider based on all signals published over the sample period. The solid, dashed, and dotted lines show the fitted curves for a 2nd order polynomial regression, where R^2 s are regressed on the percentage share of long signals. The solid line shows the estimated curve for the full sample of all signal providers, whereas the dashed (dotted) line plots the fitted curve for the subsample of signal providers with a below-median (above-median) variation in leverage. The leverage variation of a signal provider is measured as its leverage standard deviation divided by its average leverage throughout the sample period.

Figure 4 – Market Neutrality versus Trade Direction and Leverage



This figure shows how p -values for the overall F -tests of the unconstrained style regressions for signal providers vary with their percentage share of signals that imply a long position. We compute the proportion of long signals for a signal provider based on all signals published over the sample period. The solid, dashed, and dotted lines show the fitted curves for a 2nd order polynomial regression, where p -values are regressed on the percentage share of long signals. The solid line shows the estimated curve for the full sample of all signal providers, whereas the dashed (dotted) line plots the fitted curve for the subsample of signal providers with a below-median (above-median) variation in leverage. The leverage variation of a signal provider is measured as its leverage standard deviation divided by its average leverage throughout the sample period.

Appendix 2 – Tables

Table 1 – Descriptive Return Statistics and Normality

	Median											
	Signal Providers (full)	Signal Providers (lower)	Signal Providers (upper)	Mutual Funds	Non-US equities	US equities	EM equities	Euro Gov Bonds	Euro Corp Bonds	US Gov Bonds	EUR Index	Gold
Mean [%]	0.07	0.00	0.48	0.27	0.32	0.27	0.35	0.20	0.25	0.09	0.02	0.18
St. Dev. [%]	1.79	0.58	3.51	1.76	2.04	1.65	2.00	0.57	0.40	0.96	0.91	2.16
Skewness	-0.21	-0.47	-0.04	-0.48	-0.35	-0.35	-0.53	-0.03	0.54	-0.18	0.11	0.42
Kurtosis	1.98	1.96	2.14	0.29	0.16	0.06	1.29	0.46	0.77	-0.09	-0.52	-0.27
Min [%]	-4.60	-1.69	-10.79	-5.25	-5.87	-4.33	-6.42	-1.27	-0.62	-2.27	-2.00	-3.97
Max [%]	4.11	1.51	10.33	3.69	4.73	3.74	4.72	1.70	1.34	2.58	2.16	5.58
JB <i>p</i> -value	0.00 ***	0.00 ***	0.00 ***	0.29	0.52	0.54	0.03 **	0.68	0.10 *	0.86	0.78	0.43

This table reports descriptive return statistics for signal providers, mutual funds, and the eight asset class indices considered in our sample. The figures reported for signal providers and mutual funds are cross-sectional medians across the individual time-series statistics. The columns *Signal Providers (lower)* and *Signal Providers (upper)* report figures for the subsample of signal providers with a below- and above-median time-series average return, respectively. *JB p-value* is the *p*-value for the Jarque-Bera test on normality of the return distribution (see Jarque and Bera, 1980). *, **, and *** denote significance at the 10-, 5- and 1-percent-level, respectively.

Table 2 – Downside Risk of Signal Providers and Mutual Funds

	Signal Providers			Mutual Funds
	Full Sample	Lower Half	Upper Half	
VaR (0.95) [%]	2.78	0.87	4.25	2.46
CVaR (0.95) [%]	3.85	1.41	7.09	3.59
Maximum Drawdown [%]	9.60	3.57	14.25	10.29
Average Drawdown (DD) [%]	3.50	1.48	5.14	4.04
Average DD Length (weeks)	6.69	9.20	5.57	9.20
Average DD Recovery (weeks)	3.15	4.20	2.71	4.55

This table report the cross-sectional median of downside risk measures for both signal providers and mutual funds. The columns *Lower Half* and *Upper Half* report figures for the subsample of signal providers with a below- and above-median time-series average return, respectively. *VaR* and *CVaR* are value-at-risk and conditional value-at-risk at a 0.95 confidence level, respectively. *Maximum Drawdown* and *Average Drawdown* are the maximum and average peak-to-through declines over the sample period, respectively, measured in percent. *Average DD Length* and *Average DD Recovery* measure the average drawdown duration and the average time it takes a portfolio to recover from a previous drawdown (in weeks), respectively.

Table 3 – Distribution of R^2 s for Return-Based Style Regressions

	Mutual Funds			Signal Providers		
	(1)	(2)	(3)	(1)	(2)	(3)
Min	0.55	0.52	0.58	-	0.00	0.06
0.25	0.80	0.81	0.83	-	0.04	0.18
Median	0.90	0.90	0.92	0.02	0.07	0.26
0.75	0.95	0.94	0.96	0.09	0.17	0.35
Max	0.98	0.98	0.98	0.42	0.63	0.69

This table shows the distribution of R^2 s for three different style regressions, performed and reported separately for mutual funds and signal providers. Columns entitled (1) refer to the regression model defined in Equation 1 with both constraints defined in Equation 2 and 3, i.e. where both short sales and the use of leverage are prohibited. Column (2) reports R^2 s for regressions that only forbid short sales, but allow for leverage. Column (3) refers to R^2 s for a completely unconstrained regression. Negative values are indicated with a dash (“-”).

Table 4 – Descriptive Trading Statistics

Panel A: eToro			
	Long [%]	Leverage	#Trades/week
Mean	58.82	1.55	10.97
Min	23.18	0.00	1.75
Max	96.12	26.41	48.85
Std. Dev.	16.49	4.77	9.68
Panel B: ZuluTrade			
	Long [%]	Leverage	#Trades/week
Mean	47.85	3.38	14.43
Min	15.07	0.01	1.17
Max	77.54	17.96	44.27
Std. Dev.	15.14	4.18	10.49
Panel C: ayondo			
	Long [%]	Leverage	#Trades/week
Mean	58.87	9.06	18.59
Min	16.25	1.86	4.38
Max	98.95	22.21	49.52
Std. Dev.	28.62	6.77	14.57
Panel D: Currensee			
	Long [%]	Leverage	#Trades/week
Mean	47.87	1.82	11.30
Min	32.85	0.51	2.25
Max	70.59	3.65	39.06
Std. Dev.	12.85	1.14	12.87

This table reports descriptive statistics on the signal providers' trading behavior, separately for each of the four platforms in our sample. *Long [%]* corresponds to the percentage share of buy trades, measured based on all trades executed by a signal provider over the sample period. *Leverage* corresponds to the typical leverage used by a signal provider, measured as time-series average across all transactions. *#Trades/week* measures the average number of published signals per week, measured throughout the sample period.

Table 5 – Distribution of p -Values for F -Tests on Market Neutrality

	Mutual Funds		Signal Providers	
Min	0.000	***	0.000	***
0.25 quantile	0.000	***	0.016	**
Median	0.000	***	0.129	
0.75 quantile	0.000	***	0.440	
Max	0.000	***	0.986	

This table reports the distribution p -values for overall F -tests, where the null hypothesis is the joint irrelevance of the eight asset market index returns as outlined in section 3.1 to explain the time-series of a mutual funds' or signal providers' return time-series. *, **, and *** denote significance at the 10-, 5- and 1-percent-level, respectively.